Triplet Extraction from Art-historical Texts for Knowledge Graph Creation

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Recent developments in image search engines, such as *iART* (Springstein et al., 2021; Schneider et al., 2022), have assisted art historians in analyzing large image collections using different kinds of similarity. A shortcoming of these systems is the lack of clarity in their methodologies and data sources used to identify visual similarities. They reflect the biases inherent in the data, which focus on a Western canon that mirrors societal imbalances. Our approach seeks to address these limitations by incorporating data from a broader scope of art-historical literature, increasing interpretability, diversity and accuracy of these systems. *Vision-language Models (VLMs),* such as *CLIP* (*Contrastive Language-image Pre-training*; Radford et al., 2021) are integrated with *knowledge graphs* to provide users with contextual information, and to improve search accuracy and diversity by incorporating domain-specific knowledge. The *knowledge graphs* are enriched with structured knowledge from academic and institutional texts, collected to represent a range of art-historical perspectives. We implement various generative models such as REBEL (*Relation Extraction By End-to-end Language Generation*; Huguet Cabot and Navigli, 2021) , KnowGL (Knowledge Generation and Linking from Text; Rossiello et al., 2023), GoLLIE (Guideline following Large Language Model for Information Extraction, Sainz et al. 2024) and DeepKE (Deep Knowledge Extraction; Zhang et al. 2022) to extract triplets from these texts and integrate them into a knowledge graph. For the *Large Language Models (LLM)*-based methods (i.e., GoLLIE and DeepKE), we optimize the prompts for our case in order to increase the performance of extraction and to extract only relations that are relevant to us. In addition, we extract scene graphs from the images contained in the data to extract triplets from the visual information and store them in our knowledge graph.
Pan et al. (2023) provide a review of the integration of *LLMs* with *knowledge graphs* and propose a roadmap for unification. Castellano et al. (2021) suggest the use of *knowledge graphs* to improve results of automated artwork analysis. Our methodology for constructing art-historical *knowledge graphs* is inspired by Jain et al. (2022), who create domain-specific graphs for triplet extraction and automated linking of entities to *Wikidata* counterparts.

Objectives

The project's objective is to enhance the explainability of image search engine results for art historians by combining *VLMs* with the explicit semantic information of *knowledge graphs*. This integration is designed to (i) increase the accuracy and relevance of automated image retrieval through domain-specific knowledge; (ii) provide users with an interface that displays a local

scene graph and a global *knowledge graph, thereby enhancing transparency;* and (iii) promote diversity in art-historical research by using training data that includes underrepresented artists and perspectives. The local *scene graph* is designed to represent elements within each image, while the global *knowledge graph* encapsulates broader relationships and contextual insights across the entire data set. Central to our research efforts is the automated generation of these *knowledge graphs*: leveraging the text processing capabilities of generative models and neural networks for scene graph prediction, we are transforming art-historical information into triplets, which are then stored in a graph database.

Data

Our data set contains 120,000 resources from museums such as Tate Gallery [1] and Rijksmuseum , [2] as well as encyclopedic materials like Reallexikon der Deutschen Kunstgeschichte . [3] 100,000 scholarly articles are selected from academic databases such as JSTOR Constellate [4], archive.org , [5] and Wikipedia articles.

Pipeline and Current State

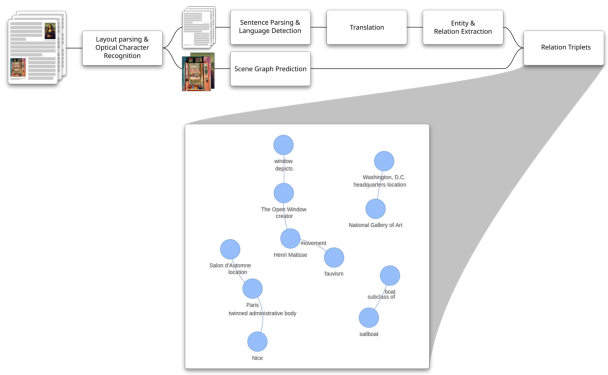

Figure 1: Knowledge graph construction pipeline. We use two methods for triplet construction: (i) Text is extracted, parsed, and translated; for triplet extraction, we use a generative model. (ii) Images detected by LayoutParser are extracted; triplets are generated using a scene graph prediction approach.

At the initial stage, our primary focus is on the extraction of triplets from textual documents. For documents in non-machine readable formats, we use *LayoutParser* (Shen et al., 2021) to isolate image and text. Given the multilingual nature of our data set, we classify the language of each sentence and then translate it into English using the NLLB-200 model (Costa-Jussà et al., 2022). Triplet prediction from these texts is performed using the afore-mentioned models *REBEL*, *KnowGL*, *GoLLIE*, and *DeepKE*. Subsequently, entity reconciliation with *Wikidata* is performed using *ReFinED* (Ayoola et al., 2022) to ensure that results are consistent with the logic of the *Wikidata* ontology.

Currently we evaluate the efficiency of these generative models in extracting triples. Due to the complexity of the scientific literature, annotation guidelines are being developed to compare human annotations with model outputs. For this, we use the semantic annotation platform *INCEpTION* (Klie et. al., 2018). These metrics are used to refine prompts for extraction of art-historically significant information. Quantitative evaluations are iteratively applied to assess the validity of triple extraction and establish a benchmark for subsequent research.

We extend the scope of our extraction to include visual content in scientific articles and websites. To achieve this, we plan to use state-of-the-art *scene graph* prediction methods, in particular *VETO* (*Vision rElation TransfOrmer;* Sudhakaran et al., 2023), to transform visual data into triplets. These triplets will be integrated into our *knowledge graph*, creating a database that encapsulates both textual and visual information. The pipeline is illustrated in Figure 1.

Outlook

Next steps are: (i) by implementing a *human-in-the-loop* approach, we intend to improve the quality of the extracted triplets; this includes engaging students and experts to review and benchmark the triplets generated by the *generative* models to ensure their correctness; (ii) by using *LLMs*, we aim to counter-check the validity of the extracted triplets (Boylan et al. 2024); (iii) We generate an annotated dataset based on the art historical datasets and manually create triplets to evaluate the pipeline; (iv) we plan to construct *knowledge graphs* on different textual data sets to compare texts from different time periods, e.g., art criticism from the 19th century with essays from the 1980s to 2000s; (v) to enhance diversity actively curating datasets that include contributions from understudied perspectives, promoting equal representation in both textual and visual analyses.

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Notes

1. https://www.tate.org.uk/ .
2. https://www.rijksmuseum.nl/en .
3. https://rdklabor.de/ .
4. https://constellate.org/ .
5. https://archive.org/ .

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